

section discusses a few open source utilities that will help to debug networking related issues.

▪ ping

We can use the *ping* command to check connectivity between hosts. It uses the Internet Control Message Protocol (ICMP) to check connectivity. Given below is the syntax of the command:

```
$ ping [ADDRESS OF HOST]
```

If you are connected to the Internet and the host is reachable, it'll start displaying *ping* statistics. Press *Ctrl+c* to abort it. We can also specify the packet count for *ping*. It'll stop automatically after sending *count* packets. For instance, the command below will stop after sending four packets:

```
$ ping -c 4 google.com
```

▪ host

This is the DNS lookup utility, which can be used to convert host names to IP addresses and vice versa. For instance, the command below prints all the IP addresses attached to the *google.com* domain:

```
$ host google.com
google.com has address 216.58.203.206
google.com has IPv6 address 2404:6800:4009:806::200e
google.com mail is handled by 30 alt2.aspmx.l.google.com.
google.com mail is handled by 10 aspmx.l.google.com.
google.com mail is handled by 50 alt4.aspmx.l.google.com.
google.com mail is handled by 20 alt1.aspmx.l.google.com.
google.com mail is handled by 40 alt3.aspmx.l.google.com.
```

Alternatively, you can also use the *nslookup* utility for DNS lookup.

▪ route

The *route* command is used to display routing table information. This table is maintained by the operating system. Execute the command given below to display the routing table on your host:

```
$ route -n
```

▪ traceroute

When we send a packet from source to destination, it may travel through multiple gateways. If we want to find those intermediate gateways, then we can use the *traceroute* command as follows:

```
$ traceroute -n google.com
```

The above command will show all the intermediate gateways between your host and *google.com*.

7) wget

Often, we download contents from the Internet/network. Most of the time, we use the browser to do this. However, GNU/Linux provides the *wget* utility, which can be used as a network downloader. This section describes a few examples. Given below is the syntax of the *wget* command:

```
wget [OPTIONS] [URL]
```

While downloading, it displays the progress bar, which shows the following:

```
-- Percentage of the download completed
-- Total amount of bytes downloaded so far
-- Current download speed
-- Remaining time to download
```

Like other utilities, *wget* is also a powerful utility. It provides various facilities to make our life easy. If your Internet connection is not stable, then downloading may be interrupted. In that case, we can provide a retry count. For instance, in the example that follows, we have provided the retry count as 3:

```
$ wget -t 3 <URL>
```

It'll retry three times before throwing out an error. To provide infinite retries, set the retry count to 0 as shown in the example below:

```
$ wget -t 0 <URL>
```

Being a very flexible utility, *wget* can also be used to restrict the downloading speed. It provides the *--limit-rate* option for this. For instance, use the command given below to set the downloading rate to 512KB:

```
$ wget --limit-rate=512K <URL>
```

One of the nice things about *wget* is that if downloading is interrupted, then it can be resumed from that point. Use the command given below to resume an interrupted download:

```
$ wget -c <URL>
```

8) Working with a remote host

We often interact with remote hosts to download or upload content. This section discusses command line utilities that will perform these tasks.

▪ scp

One of the common tasks is to transfer files between the remote and the local host. GNU/Linux provides a remote